Week 2 Exercises

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Please complete all exercises below. You may use stringr, lubridate, or the forcats library.

```
library(stringr)
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.2.3
```

```
##
## Attaching package: 'lubridate'
```

```
## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(forcats)
```

```
## Warning: package 'forcats' was built under R version 4.2.3
```

Exercise 1

Read the sales_pipe.txt file into an R data frame as sales.

```
sales <- read.delim("C:/Users/Nischal/Downloads/sales_pipe.txt",
                    header = TRUE,
                    check.names = FALSE,
                    stringsAsFactors=FALSE,
                    sep = "|")
```

Exercise 2

You can extract a vector of columns names from a data frame using the colnames() function. Notice the first column has some odd characters. Change the column name for the FIRST column in the sales date frame to Row.ID.

Note: You will need to assign the first element of colnames to a single character.

```
#first find the names of the column headers
colnames(sales)
```

```
##  [1] "\xef..Row.ID"  "Order.ID"      "Order.Date"    "Ship.Date"
##  [5] "Ship.Mode"     "Customer.ID"   "Customer.Name" "Segment"
##  [9] "Country"       "City"          "State"         "Postal.Code"
## [13] "Region"        "Product.ID"    "Category"      "Sub.Category"
## [17] "Product.Name"  "Sales"         "Quantity"      "Discount"
## [21] "Profit"
```

```r
#it shows that one of them has special characters, now lets change that to "Row.ID"
colnames(sales)[colnames(sales) == "\xef..Row.ID"] <- "Row.ID"

#check the names again to check if it has now been changed. It has.
colnames(sales)
```

```
##  [1] "Row.ID"        "Order.ID"      "Order.Date"    "Ship.Date"
##  [5] "Ship.Mode"     "Customer.ID"   "Customer.Name" "Segment"
##  [9] "Country"       "City"          "State"         "Postal.Code"
## [13] "Region"        "Product.ID"    "Category"      "Sub.Category"
## [17] "Product.Name"  "Sales"         "Quantity"      "Discount"
## [21] "Profit"
```

Exercise 3

Convert both Ship.Date and Order.Date to date vectors within the sales data frame. What is the number of days between the most recent order and the oldest order? How many years is that? How many weeks?

Note: Use lubridate

```r
#first lets use lubridate's mdy function to change all the date format and plug them back into the sale
sales$Ship.Date  <- mdy(sales$Ship.Date)
sales$Order.Date <- mdy(sales$Order.Date)

#summary to give details about the Order.date variable of the sales data set
summary(sales$Order.Date)
```

```
##         Min.      1st Qu.       Median         Mean      3rd Qu.         Max.
## "2014-01-03" "2015-06-11" "2016-07-17" "2016-05-09" "2017-05-28" "2017-12-30"
```

```r
#now lets extract that variable and its cells and assign them to variable dates
dates <- ymd(sales$Order.Date)

#using the intervals function lets assign them to date_intervals to calculate how many days are between
date_interval <- interval(min(dates), max(dates))

#because the data is in date format, change them into days.
num_days <- as.integer(date_interval / ddays(1))

#number of days
num_days
```

```
## [1] 1457
```

```r
#calculation for number of years
num_days/365
```

```
## [1] 3.991781
```

```r
#calculation for number of weeks
num_days/7
```

```
## [1] 208.1429
```

Exercise 4

What is the average number of days it takes to ship an order?

```r
#for this we need number of days between the order.date variable and ship.date variable. Lets assign th

subtracted_values <- sales$Ship.Date - sales$Order.Date
mean(subtracted_values)
```

```
## Time difference of 3.908482 days
```

Exercise 5

How many customers have the first name Bill? You will need to split the customer name into first and last name segments and then use a regular expression to match the first name bill. Use the length() function to determine the number of customers with the first name Bill in the sales data.

```r
#lets first create a matrix with the str_split function to split the name string into two cells separat
name <- str_split_fixed(sales$Customer.Name, " ", n = 2)


#now using which, we can find how many indices exist where the first name is Bill and the length functi
length(which(name[, 1] == "Bill"))
```

```
## [1] 37
```

Exercise 6

How many mentions of the word 'table' are there in the Product.Name column? **Note you can do this in one line of code**

```r
#we can use the str_count to find how many times table shows up in the sales data set

sum(str_count(sales$Product.Name, fixed('table')))
```

```
## [1] 240
```

Exercise 7

Create a table of counts for each state in the sales data. The counts table should be ordered alphabetically from A to Z.

```r
#the code I originally wrote was concise and only 2 steps but failed to sort them alphabetically. They

# first lets create a table with the states in sales data set
states <- table(sales$State)

# Sort the counts table alphabetically
sorted_state <- sort(states)

# now we create a data frame with state and count columns
states_final <- data.frame(State = names(sorted_state), Count = sorted_state)

# now we organize them again by order
states_final <- states_final[order(states_final$State), ]

states_final
```

```
##                  State           Count.Var1 Count.Freq
```

```
## 21              Alabama              Alabama         28
## 40              Arizona              Arizona        119
## 15             Arkansas             Arkansas         22
## 49           California           California        993
## 38             Colorado             Colorado         90
## 28          Connecticut          Connecticut         50
## 27             Delaware             Delaware         47
## 1  District of Columbia District of Columbia          1
## 42              Florida              Florida        186
## 35              Georgia              Georgia         79
## 7                 Idaho                Idaho          9
## 45              Illinois             Illinois        286
## 34              Indiana              Indiana         74
## 11                 Iowa                 Iowa         11
## 13               Kansas               Kansas         16
## 32              Kentucky             Kentucky         64
## 14             Louisiana            Louisiana         18
## 4                 Maine                Maine          4
## 31              Maryland             Maryland         63
## 33         Massachusetts        Massachusetts         71
## 41              Michigan             Michigan        142
## 26             Minnesota            Minnesota         41
## 19           Mississippi          Mississippi         27
## 23              Missouri             Missouri         37
## 3                Montana              Montana          2
## 18              Nebraska             Nebraska         26
## 16                Nevada               Nevada         24
## 8          New Hampshire        New Hampshire          9
## 30            New Jersey           New Jersey         58
## 12            New Mexico           New Mexico         11
## 48              New York             New York        555
## 39        North Carolina       North Carolina        117
## 6           North Dakota         North Dakota          7
## 43                  Ohio                 Ohio        211
## 24              Oklahoma             Oklahoma         38
## 29                Oregon               Oregon         56
## 46          Pennsylvania         Pennsylvania        312
## 17          Rhode Island         Rhode Island         25
## 22        South Carolina       South Carolina         28
## 9           South Dakota         South Dakota          9
## 37             Tennessee            Tennessee         88
## 47                 Texas                Texas        460
## 20                  Utah                 Utah         27
## 10               Vermont              Vermont         10
## 36              Virginia             Virginia         80
## 44            Washington           Washington        254
## 5          West Virginia        West Virginia          4
## 25             Wisconsin            Wisconsin         38
## 2                Wyoming              Wyoming          1
```

Exercise 8

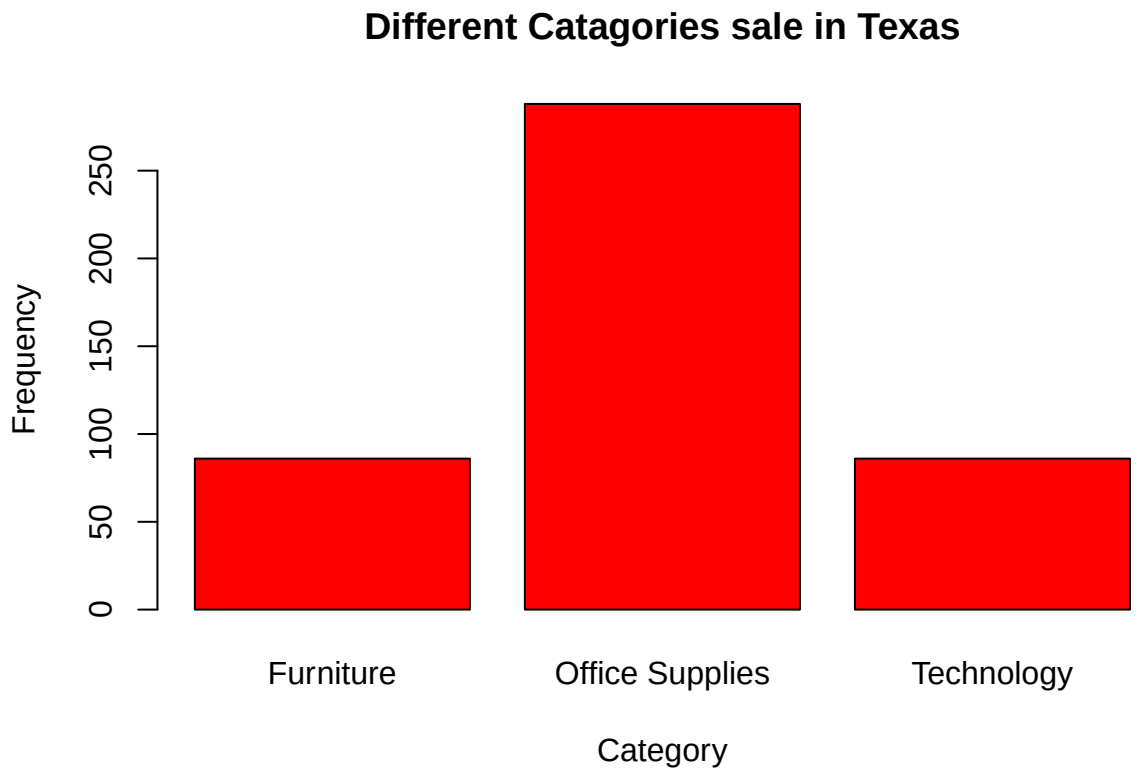
Create an alphabetically ordered barplot for each sales Category in the State of Texas.

```r
#lets first extract sales of Texas from the sales data set
sales_texas <- sales[sales$State == 'Texas',]

#now lets extract the category variable of that table
category_freq <- table(sales_texas$Category)

#we can now order them alphabetically
category_freq <- category_freq[order(names(category_freq))]

#we can now use the bar plot function to create a bar plot of the data,
barplot(category_freq,
        main = "Different Catagories sale in Texas",
        xlab = "Category",
        ylab = "Frequency",
        col = "RED",
        border = "black")
```



Exercise 9

Find the average profit by region. **Note: You will need to use the aggregate() function to do this. To understand how the function works type ?aggregate in the console.**

```r
#this code aggregates the profit depending on the region and FUN (= function) which is sum in this case

prof <- aggregate(Profit ~ Region, data = sales, FUN = sum, na.action = na.omit)
prof
```

```
##    Region    Profit
## 1 Central 24254.845
## 2    East 42485.498
## 3   South  8311.297
## 4    West 51973.214
```

Exercise 10

Find the average profit by order year. **Note: You will need to use the aggregate() function to do this. To understand how the function works type ?aggregate in the console.**

```
# Your code here
#first lets convert the order.date variable in sale data into year
sales$Order.Date <- year(sales$Order.Date)

#this code aggregates the profit depending on the years and FUN (= function) which is sum in this case.
dateprof <- aggregate(Profit ~ Order.Date, data = sales, FUN = sum, na.action = na.omit)
dateprof
```

```
##   Order.Date   Profit
## 1       2014 31052.73
## 2       2015 21824.21
## 3       2016 38269.30
## 4       2017 35878.62
```